

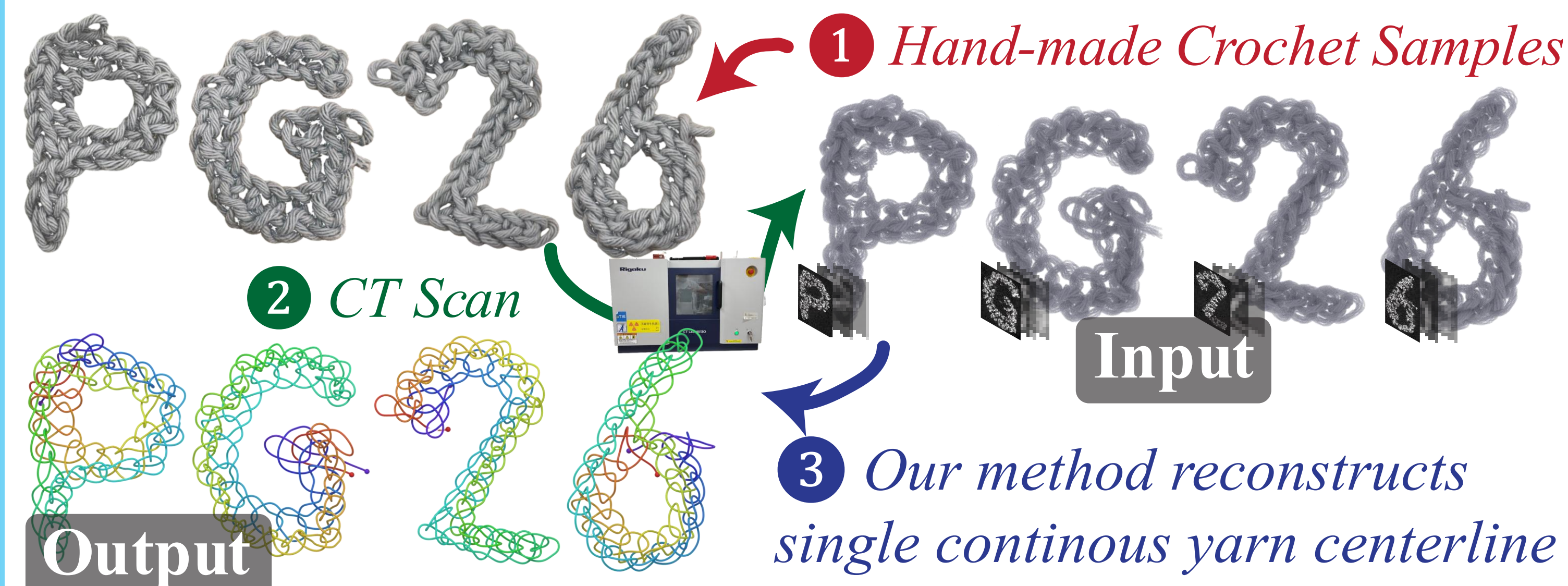
CT2Yarn: Yarn Level Reconstruction of Crochet from Computed Tomography

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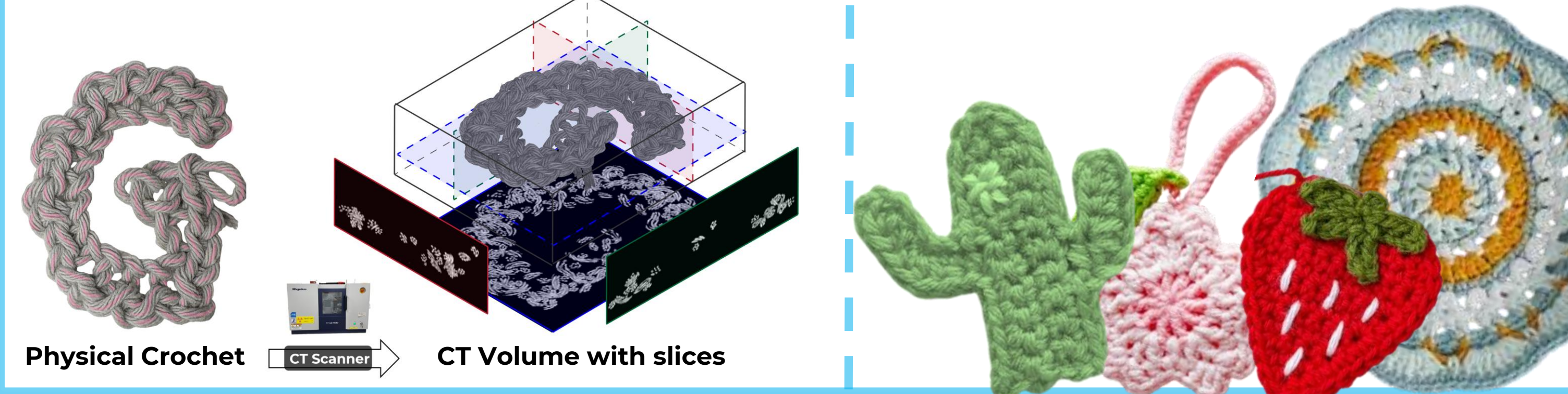
The University of Tokyo

1. Introduction

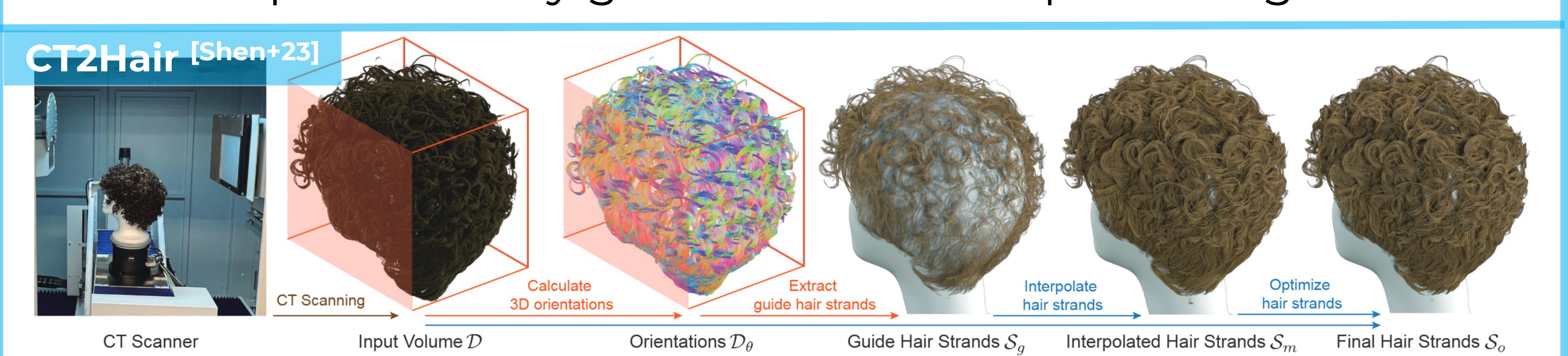
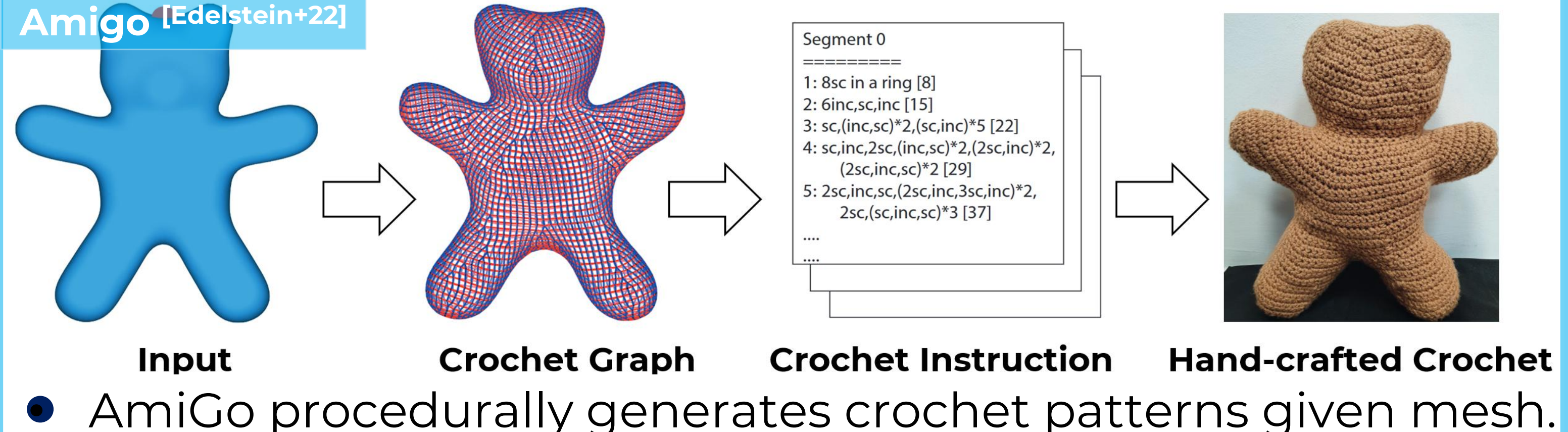
- Human-in-the-loop** method recovers crochet centerline.



- Micro-CT** reveals the hidden internal structure.

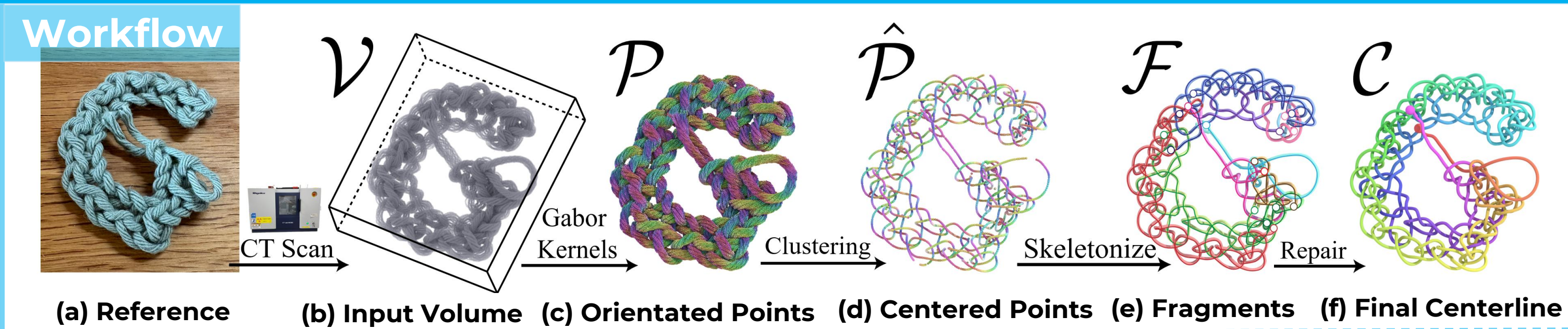


2. Related works

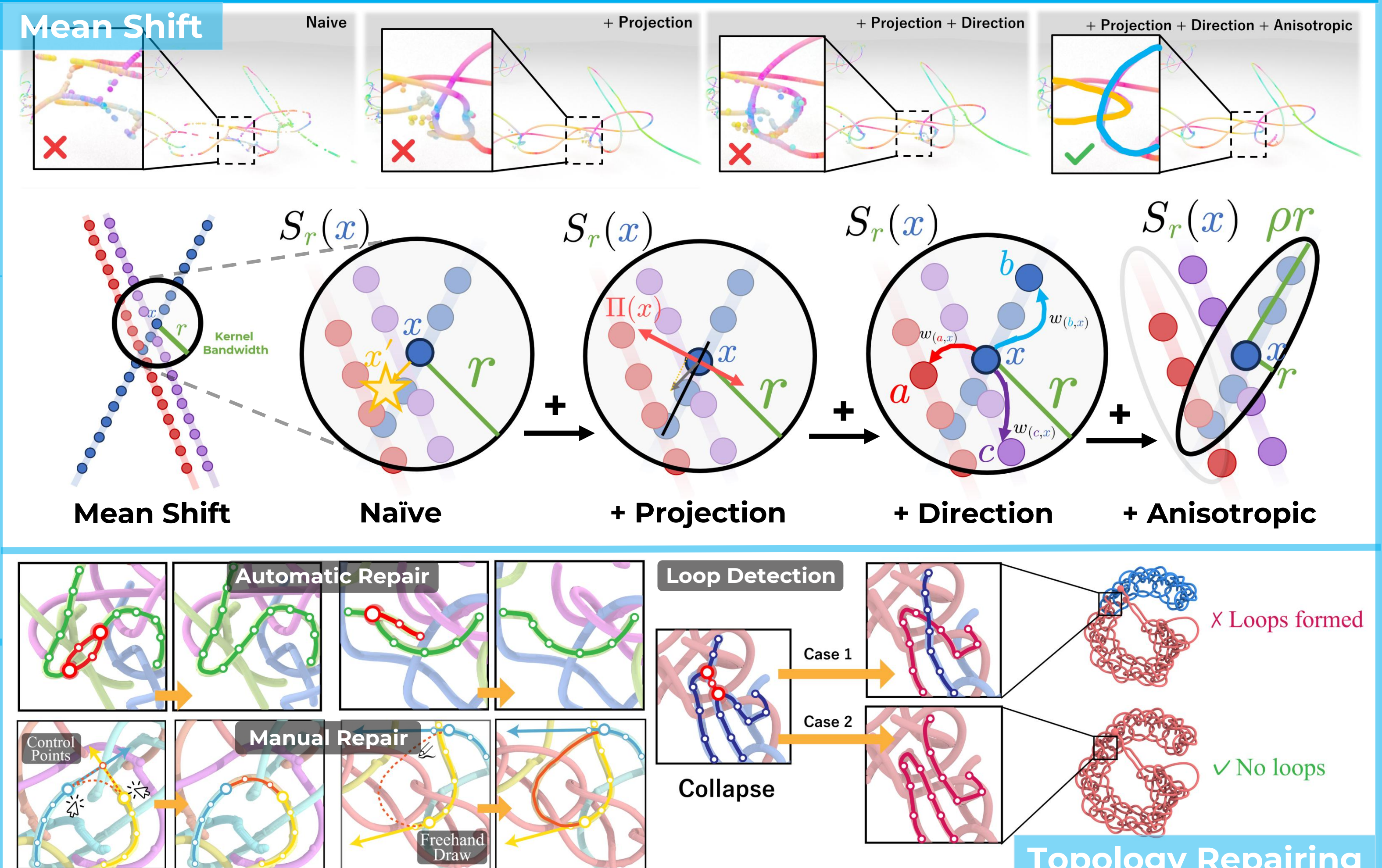
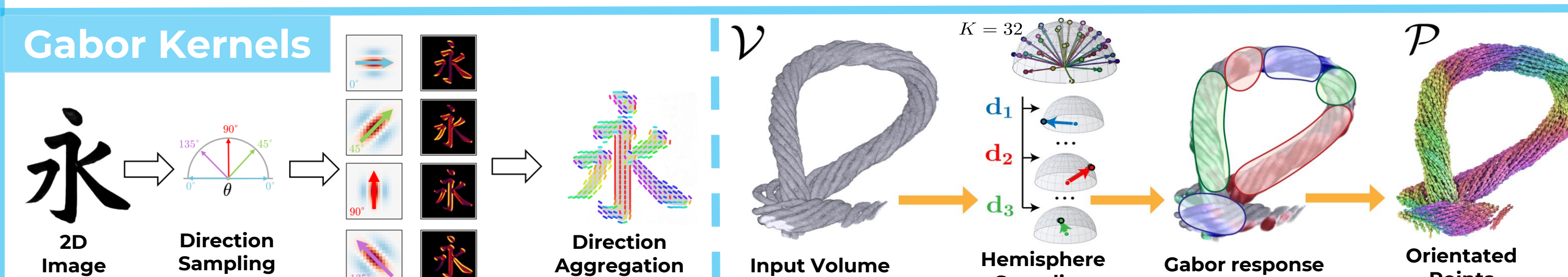


- CT2Hair** reconstructs independent hair strands with wig scan.
- But, crochet has **Hierarchy**!
- Wrong connections in local** brings error in **global topology**!

3. Methodology

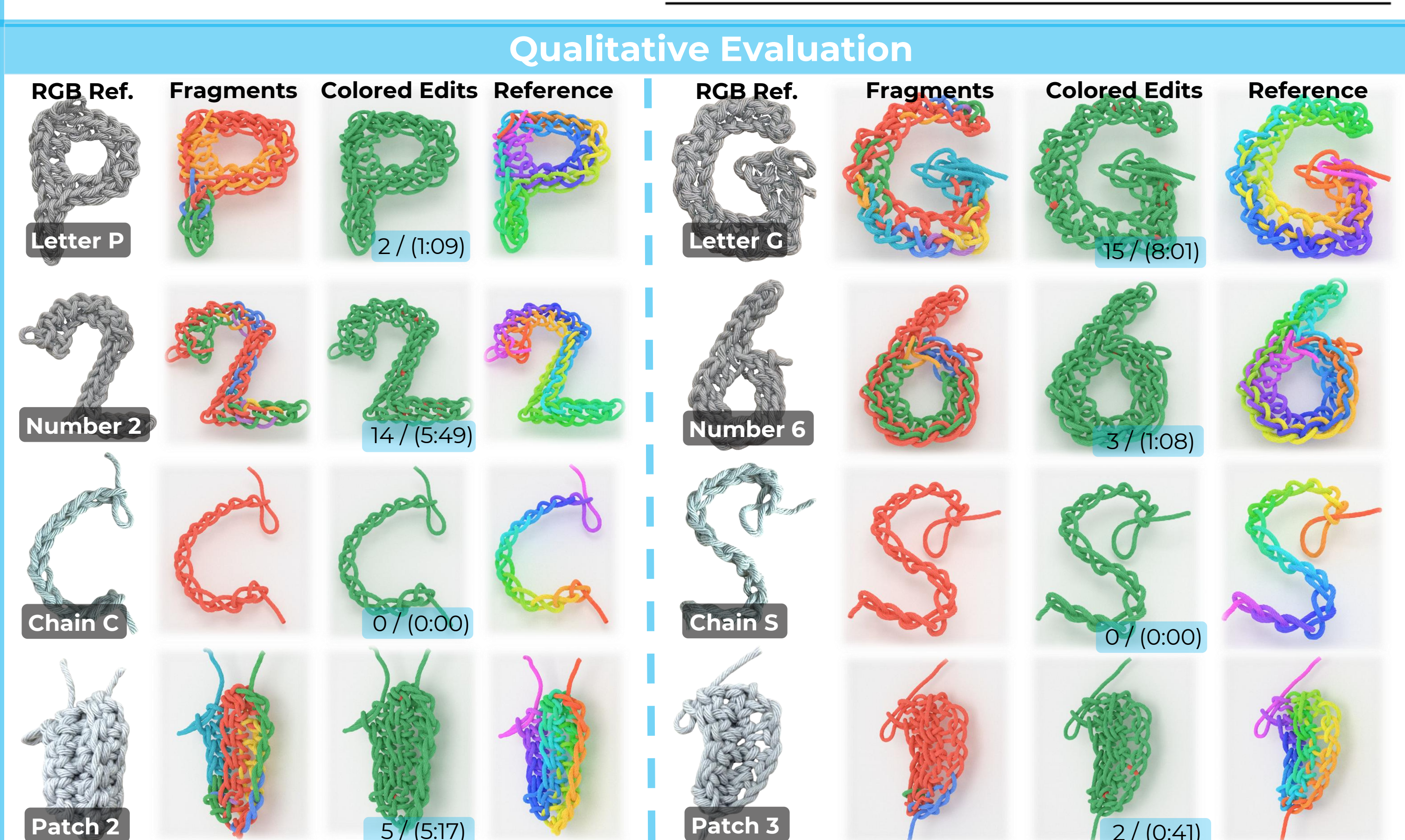


- Orientation estimation** Gabor extracts **local fiber direction**.
- Center-point extraction**. Anisotropic mean-shift aggregates directions at the yarn scale. **Perpendicular projection**, **direction-aware weighting**, and **ellipsoid kernels** concentrate points toward the yarn centerline.
- Topology reconstruction**. MST connects fragments. **Topology repair** resolve common breaks with **UI repair**.

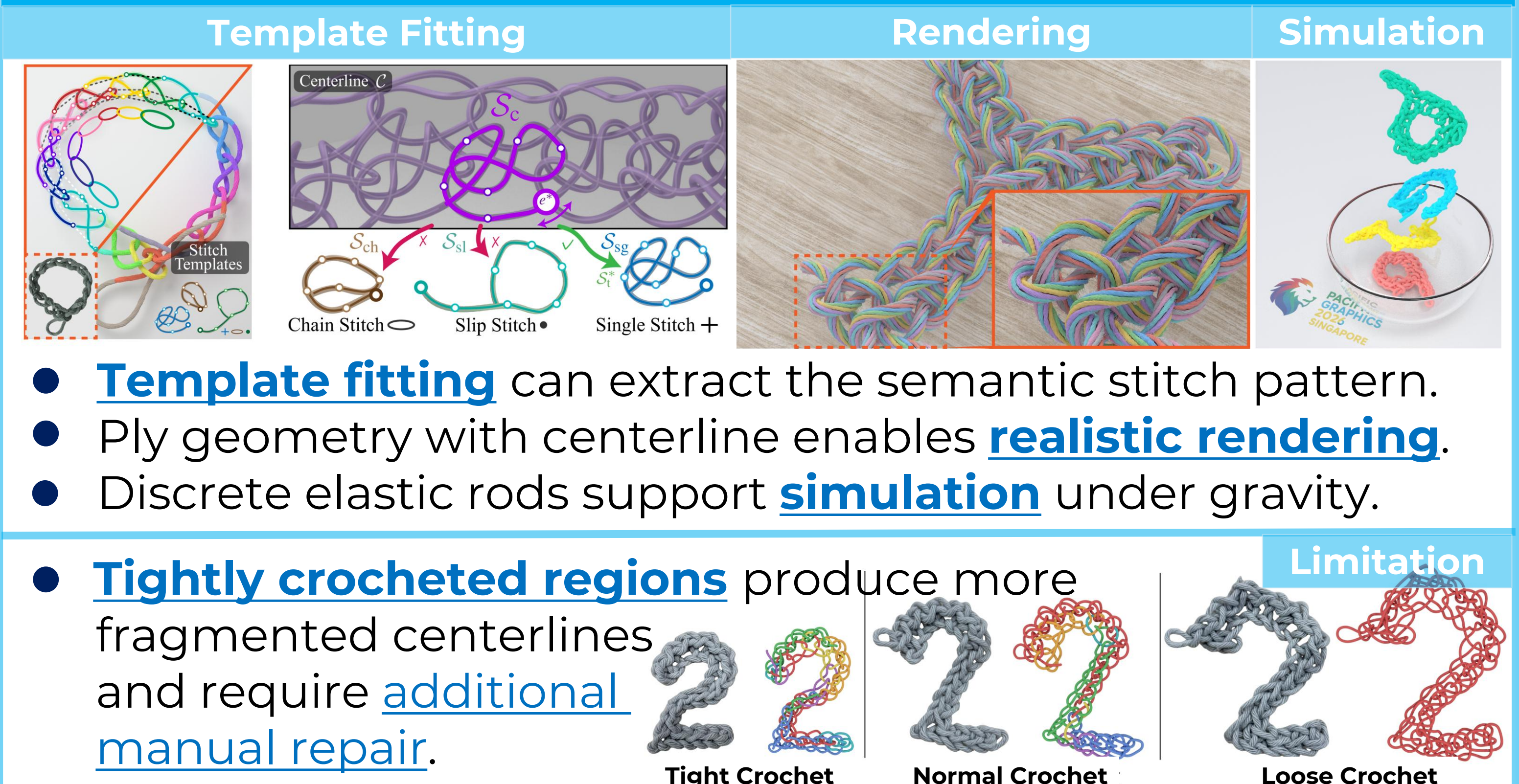


4. Results

Quantitative Evaluation		Reference-free		Reference-based		Effort	
Method	#C	$\bar{L}_c$	$E_L \downarrow$	$\bar{\alpha} \downarrow$	$\ell_{CD} \downarrow$	$\ell_2 \downarrow$	edits time
CT2Hair (Gab.)	88	314	0.115	10.55	16.99	18.55	— —
CT2Hair (Grad.)	115	236	0.271	12.75	20.60	24.67	— —
Ours (Auto)	15	3814	0.002	2.66	2.43	2.26	— —
Ours (Manual)	1	30359	—	—	—	—	4 2:46



5. Applications and Limitations



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References:

- Shen, Y., Saito, S., Wang, Z., Maury, O., Wu, C., Hodgins, J., ... & Nam, G. (2023). Ct2hair: High-fidelity 3d hair modeling using computed tomography. *ACM Transactions on Graphics (TOG)*, 42(4), 1-13.
- Edelstein, M., Peleg, H., Itzhaky, S., & Ben-Chen, M. (2022, October). Amigo: Computational design of amigurumi crochet patterns. In *Proceedings of the 7th Annual ACM Symposium on Computational Fabrication* (pp. 1-11).



Project Page